

Module 4: Cross-ecosystem contrast comparison

Instructional time: 30 min | Bloom: Evaluate | Notebook: notebooks/04_cross_ecosystem.ipynb

1. Overview

Module 4 compares independently generated lung transcriptomic contrasts after methods calibration. Native expression levels are shown only to demonstrate incomparability. What can travel is within-resource log fold-change and the pathway **normalized enrichment score (NES)**, the per-pathway statistic gene set enrichment analysis returns after correcting for gene-set size. The headline analysis is **shared-mechanism concordance** against chance; global NES Spearman is supporting context only.

Key components. Four within-resource contrasts (GEO **idiopathic pulmonary fibrosis (IPF)**, GTEx aging, OSD-248 spaceflight, HuBMAP aging illustration); a HuBMAP to GTEx level bridge; pathway NES comparison across contrast pairs; and shared-mechanism concordance tested against permutation nulls. Calibration checks on the flight contrast establish that the pipeline is sound before any biology is read from it.

Learning objectives. After this module you will be able to:

- Explain why expression levels produced by independent pipelines cannot be merged, and identify which quantities can be compared across resources instead.
- Judge whether a contrast is fit to interpret, using spike-in calibration and agreement with an independently produced differential-expression table as the evidence.
- Test whether pathway programs agree in direction beyond chance using a permutation null, and explain why a global correlation across all pathways cannot answer the same question.
- Distinguish a null result from an uninterpretable one, and state which of the two a given contrast produced and what would be needed to resolve it.

No CFDE program supplies fibrotic human lung; the disease arm is external GEO GSE150910 (NIH, not CFDE).

2. Before you begin

Prerequisites. Modules 1 to 3 complete; conda env `CFDE_lung_env` (or the pinned pip venv); Module 2 integrated object and pseudobulk tables present.

Time. Instructional: ~30 min. Compute (measured from `module4_run_params.json`, sensitivity gated off): notebook wall ~5 min / ~301 s compute; peak RSS ~5.5 GB. Course RAM: 16 GB recommended, 8 GB minimum (Module 2 DE peaks at ~7.6 GB). Free disk: several GB for tables/figures.

Data required.

Run order. Run the analysis sections in order; the notebook carries state between cells.

Biological question

When disease, aging, and spaceflight lung contrasts are each analyzed in their own resource, which pathway programs agree in direction beyond chance, and which contrasts are uninterpretable given the data?

Spaceflight belongs in that comparison rather than sitting beside it as an unrelated third condition. The combination of microgravity and space radiation is studied explicitly as a model of accelerated aging, producing the inflammation, cellular senescence and fibrotic remodeling signatures that characterize aged tissue, and a genomic radiation-age association derived for spaceflight has already been carried into lung disease (references 6 and 7). That is what makes a flight contrast worth testing against a chronological aging contrast and against a fibrotic disease contrast: if the accelerated aging framing holds at the pathway level, the three should share directional programs. Whether they do in these particular datasets is the empirical question this module answers.

Data used here

Resource	Accession	Contrast	Local role
GEO	GSE150910	H_DISEASE IPF vs control	Bulk counts
GTEX v10 lung	gene_reads / gene_tpm	H_AGING_GTEX older vs younger	Bulk counts + TPM bridge
NASA OSDR	OSD-248	M_FLIGHT ISS-T flight vs GC	Mouse bulk counts (+ ERCC)
HuBMAP	teaching donors	H_AGING_HUBMAP older vs younger	2 vs 2 illustration
Orthologs	Ensembl one2one	mouse to human symbols	data/reference/
Gene sets	Hallmark / Reactome / GO BP	NES	data/genesets/

Module 1 introduced the download map; Module 2 supplied composition-weighted HuBMAP pseudobulk for the GTEX bridge. Sample sizes:

resource	contrast_id	arm	n	notes
GSE150910	H_DISEASE	IPF	103	CHP excluded from teaching contrast
GSE150910	H_DISEASE	control	103	CHP excluded from teaching contrast
OSD-248	M_FLIGHT	ground	10	ISS-T ~60 Carcass; duration-matched GC (not hindlimb unloading)
OSD-248	M_FLIGHT	flight	10	ISS-T ~60 Carcass; duration-matched GC (not hindlimb unloading)
OSD-248	M_FLIGHT	flight_qa_below_threshold	1	Mmus_C57-6T_LNG_FLT_ISS-T_Rep7_F7
GTEX	H_AGING_GTEX	younger_brackets	100	brackets=['20-29', '30-39']; samples=257; DE uses one sample/subject capped at 40/arm
GTEX	H_AGING_GTEX	older_brackets	269	brackets=['60-69', '70-79']; samples=655; DE uses one sample/subject capped at 40/arm
HuBMAP	H_AGING_HUBMAP	older	2	illustration only; donors=Donor_3,Donor_4; ages=56.8,52.96

HuBMAP	H_AGING_HUBMAP	younger	2	illustration only; donors=Donor_1,Donor_2; ages=37.0,25.0
HuBMAP	bridge	composition_weighted_genes	42258	linear-scale pseudobulk from Module 2 (levels only)
GTEX	bridge	lung_mean_tpm_genes	59033	mean TPM for levels/bridge only; age DE uses gene_reads

Source: module4_sample_counts.tsv

3. Methods

Parameters below are live values under `module4` in `config/paths.yaml`. Scripts live in `scripts/nb06_crossres/`. Native expression levels never travel across ecosystems; within-resource log fold-change and pathway NES do. The headline analysis is **shared-mechanism concordance** against chance; global NES Spearman is supporting context only.

Each Methods step below corresponds to a section of `notebooks/04_cross_ecosystem.ipynb`. The notebook is ordered by the question each result answers rather than by the order the steps run, so the two sequences do not track each other:

Methods step	Notebook section
1. Within-resource DE (pydeseq2)	2. Four within-resource contrasts and calibration
2. Calibration of the flight contrast	2. Four within-resource contrasts and calibration
3. Ortholog mapping (mouse to human)	3. Ortholog mapping and pathway concordance
4. HuBMAP-GTEX level bridge	4. HuBMAP to GTEX level bridge
5. Pathway NES and configured pairs	3. Ortholog mapping and pathway concordance
6. Shared-mechanism concordance	3. Ortholog mapping and pathway concordance

The notebook runs in the order the questions are asked rather than the order the steps execute, so step 4 appears after steps 5 and 6. Notebook section 1, on whether native expression levels can be compared across resources, is reported under Results as "Native levels do not travel" rather than as a Methods step.

1. Within-resource DE (pydeseq2)

Script. `scripts/nb06_crossres/contrasts.py` (`contrast_h_disease`, `contrast_h_aging_gtex`, `contrast_h_aging_hubmap`, `contrast_m_flight`, `_pydeseq2_contrast`); loaders in `scripts/nb06_crossres/load.py` and `scripts/nb06_crossres/hubmap_donor_counts.py`.

What. Fit negative binomial DE separately in each resource with explicit reference levels (`Treatment(reference)`). Never concatenate counts across studies or species. Four contrasts only:

- `H_DISEASE`: GEO GSE150910 IPF vs control (~ `condition`)
- `H_AGING_GTEX`: GTEX v10 lung older vs younger brackets
- `H_AGING_HUBMAP`: HuBMAP teaching donors older vs younger (2 vs 2 illustration from summed counts)
- `M_FLIGHT`: OSD-248 ISS-T flight vs ground control

Filter genes with low total counts (`de_min_count_sum`). Round RSEM expected counts to integers before `pydeseq2` where required.

Parameters: see Appendix, section 1.

Why. Units differ (TPM, rounded sums, RSEM counts). Alphabetical factor orders would invert several contrasts; references are set deliberately. HuBMAP aging is illustration-only: ages sit inside GTEx middle brackets and power differs by an order of magnitude from `H_AGING_GTEX`.

2. Calibration of the flight contrast

Script. `scripts/nb06_crossres/ercc_calib.py` (`ercc_calibration_frame`, `ercc_calibration_summary`, `plot_ercc_calibration`); GeneLab subset in `scripts/nb06_crossres/genelab_subset.py`; size-factor policy in `contrast_m_flight`.

What. ExFold spike-in calibration for OSD-248 (flight Mix1 vs GC Mix2) against Thermo Fisher expected fold changes. Cross-check endogenous logFC against a committed GeneLab ISS-T DE subset (`osd248_genelab_isst_de_subset`). Estimate size factors on endogenous genes only (`module4.osd248.exclude_ercc_from_size_factors: true`) while keeping ERCC rows for calibration plots. Optional RIN/QA sensitivity uses `rin_sensitivity` / `rin_sensitivity_threshold` 6.0.

Parameters: see Appendix, section 2.

Why. Spike-ins can dominate ranks when mixes differ by arm. Calibration asks whether the flight contrast is methodologically usable before biology claims. Putting ERCC back into size factors was tested and rejected (ground ERCC fraction was higher; pathway `p_up` stayed 0). Adult globin share was checked and was not a material flight-specific driver: 1.33 percent of the non-ERCC library in flight against 0.96 percent in ground, a difference of 0.36 percentage points. Even when methods checks pass, flight pathway passers that are unanimously down remain **uninterpretable, not null**, for cross-ecosystem concordance.

3. Ortholog mapping (mouse to human)

Script. `scripts/nb06_crossres/orthologs.py` (`load_ortholog_table`, `ortholog_loss_record`, `align_logfc_by_ortholog`).

What. Map mouse genes to human symbols with Ensembl one-to-one orthologs (`data/reference/human_mouse_orthologs_one2one.tsv`; BioMart download metadata under `module4.orthologs_meta`). Record loss in `module4_ortholog_loss.tsv`. Leading-edge overlaps for cross-species mechanism pairs restrict both sides to the mappable human-symbol subspace.

Parameters: see Appendix, section 3.

Why. Many-to-many orthologs would double-count. Loss is part of the result and must travel with any human-space NES comparison that includes mouse flight.

4. HuBMAP-GTEx level bridge (demonstration only)

Script. `scripts/nb06_crossres/compare.py` (`hubmap_gtex_bridge_table`, `plot_pseudobulk_vs_gtex`); inputs from Module 2 pseudobulk exports and GTEx TPM means via `scripts/nb06_crossres/load.py`.

What. Compare Module 2 composition-weighted and max-across-labels HuBMAP pseudobulk vectors to GTEx lung mean TPM on shared symbols. This demonstrates that native levels do not travel; it is not a DE input. Production runs favor composition-weighted over max-across for bulk-like tracking.

Parameters: see Appendix, section 4.

Why. Showing incomparability of native levels is the pedagogical point before NES concordance. Levels never enter the headline mechanism test.

5. Pathway NES and configured pairs

Script. `scripts/nb06_crossres/compare.py` (`prerank_contrast`, `nes_comparison`, `compare_configured_nes_pairs`, `resolve_nes_sensitivity`).

What. Prerank GSEA on each contrast's ranked logFC against local Hallmark + Reactome GMTs (no KEGG; GO BP is sensitivity-only). Prerank defaults: `min_size=15`, `max_size=500`, `permutation_num=100`. Compare exactly four NES pairs from `module4.nes_pairs` (`H_AGING_HUBMAP` only vs `H_AGING_GTEX`). Global Spearman on overlapping gene sets is recorded as supporting context with an independence caveat; it is not the headline. GO BP sensitivity prerank is gated: `module4.sensitivity.enabled: false` loads the committed TSV (`module4_nes_sensitivity.tsv`) to keep RAM near ~5.5 GB instead of recomputing (~12 GB peak).

Parameters: see Appendix, section 5.

Why. Global Spearman is not independent evidence across overlapping gene sets. Mechanism concordance (next step) uses pass/fail and sign agreement with permutation nulls and is the correct headline object.

6. Shared-mechanism concordance (headline)

Script. `scripts/nb06_crossres/mechanisms.py` (`run_mechanism_analysis`, `pass_mask`, `permute_concordant_count`, `analytic_null`, `leading_edge_overlap`, `cluster_mechanisms`).

What. Define pathway passers with $FDR\ q < 0.25$ and absolute NES ≥ 1.5 in each contrast; concordant pathways agree in sign. Compare observed concordant counts to analytic and permutation nulls (`n_permutations=2000`, `random_seed=0`) that preserve observed marginals (including flight's all-down pathway passers). All pathways ship as **tier 3** (concordance only): leading-edge Jaccard on this run (~0.21-0.48) has no natural tier 1/2 cut. Suspect OxPhos / translation terms are flagged, never auto-excluded. Cross-species leading edges restrict to ortholog-mappable human symbols. Flight-involving pairs with unanimous down passers are labeled **uninterpretable, not null**.

Parameters: see Appendix, section 6.

Why. Permutation preserves polarity marginals so enrichment is not automatic from "everything is down" alone. Tier 3-only labeling prevents overclaiming shared leading-edge mechanisms when Jaccard stays moderate. The clustering result is binding on the prose: where it declines to merge, the pathways are reported separately. Disease-aging concordance does not rise above the null; ECM direction inferred from global rho is corrected by the pathway screen; aging-flight exceeds permutation while remaining biologically suspect.

4. Results

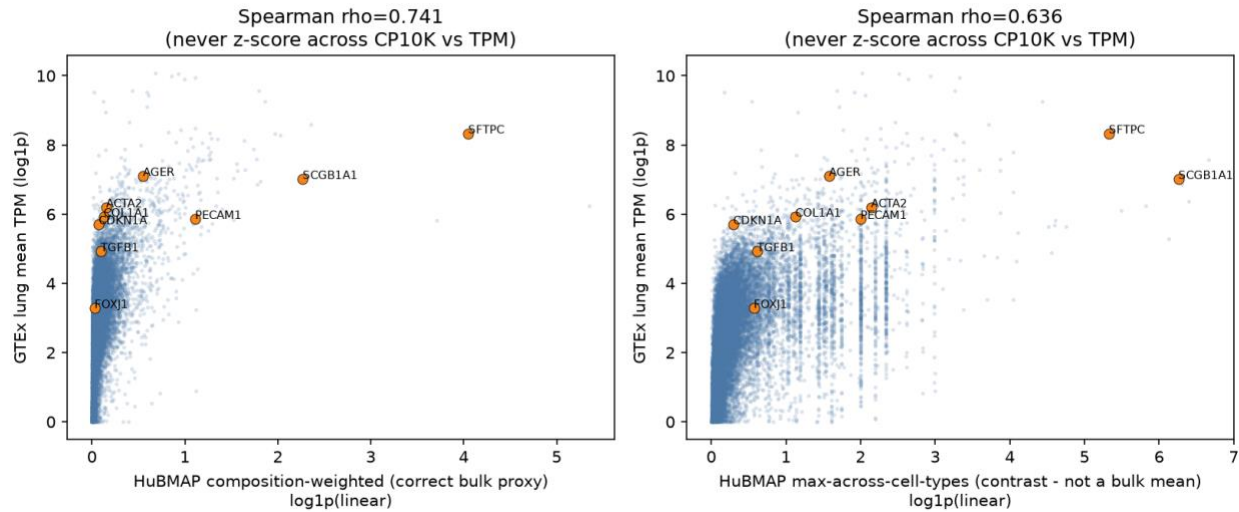
Ortholog loss

metric	n
genes_in_human_contrast	18401
genes_in_mouse_contrast	15429
one2one_orthologs_available	17041
retained_after_intersection	13577
dropped_human_side	4824
dropped_mouse_side	1852

Source: `module4_ortholog_loss.tsv`

Methods checks that pass

HuBMAP vs GTEx: composition-weighted tracks bulk more closely ($\rho = 0.741$ against 0.636); max-across is a contrast, not a bulk mean



HuBMAP pseudobulk versus GTEx lung mean TPM bridge

ERCC calibration on the production run: slope 0.98884, R^2 0.8817. The flight contrast computed here also agrees with GeneLab's independently produced differential expression across shared symbols, which is what that cross-check is for: two pipelines run on the same samples should rank genes the same way, and a disagreement would indicate a processing error rather than biology. The HuBMAP-GTEx bridge favors composition-weighted over max-across vectors: Spearman ρ 0.741 against 0.636 across 18,055 shared genes.

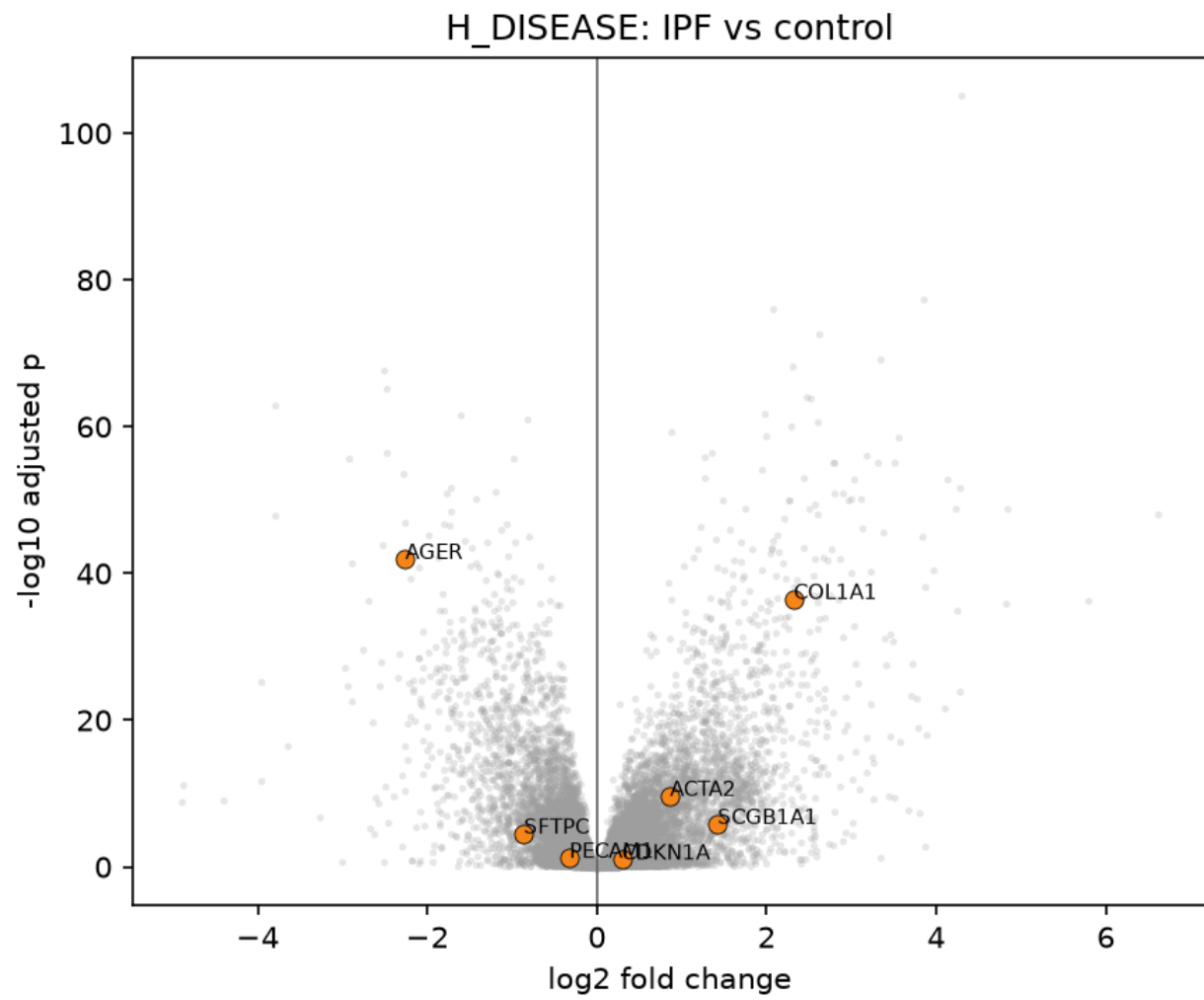
Native levels do not travel

gene_symbol_human	gtex_lung_mean_tpm	gse150910_mean_counts_ipf	osd248_mean_counts_flight	units_note
SFTPC	4156.258313	627510.165049	280904.5	TPM vs linear CP10K- pseudobulk vs raw counts; human vs mouse -- not comparable
AGER	1214.833166	4364.708738	29046.4	TPM vs linear CP10K- pseudobulk vs raw counts; human vs mouse -- not comparable
SCGB1A1	1113.153419	426.776699	698380.1	TPM vs linear CP10K- pseudobulk vs raw counts; human vs mouse -- not comparable
FOXJ1	25.705356		1519.6	TPM vs linear CP10K-

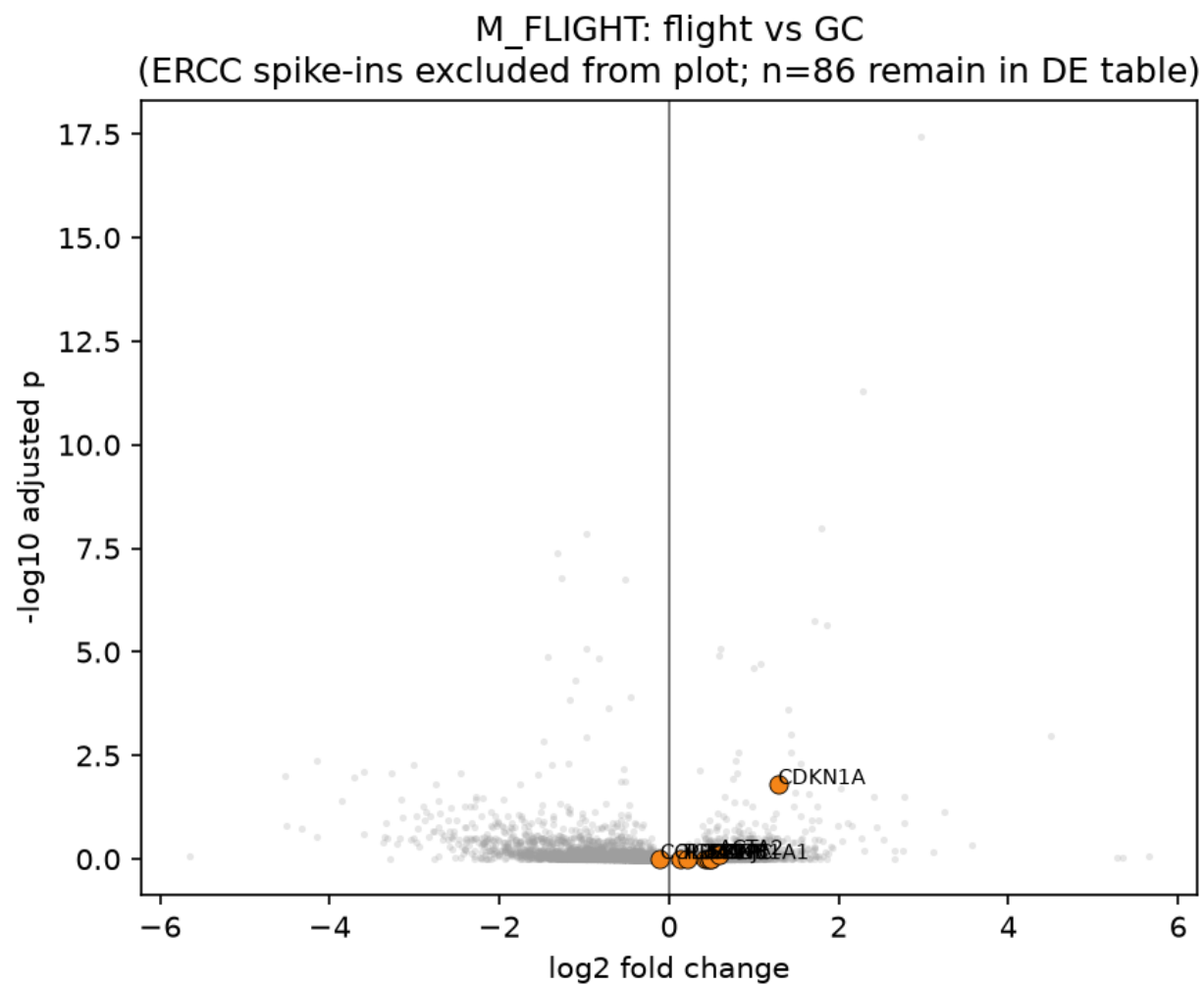
				pseudobulk vs raw counts; human vs mouse -- not comparable
PECAM1	350.185872	1236.68932	24226.5	TPM vs linear CP10K- pseudobulk vs raw counts; human vs mouse -- not comparable
COL1A1	374.837595	20970.009709	3038.4	TPM vs linear CP10K- pseudobulk vs raw counts; human vs mouse -- not comparable
ACTA2	492.614129	35195.407767	6660.552	TPM vs linear CP10K- pseudobulk vs raw counts; human vs mouse -- not comparable
CDKN1A	301.076699	10327.038835	2325.2	TPM vs linear CP10K- pseudobulk vs raw counts; human vs mouse -- not comparable
TGFB1	136.778551		944.8	TPM vs linear CP10K- pseudobulk vs raw counts; human vs mouse -- not comparable

5 further columns in the source file. Source: module4_native_levels_panel.tsv

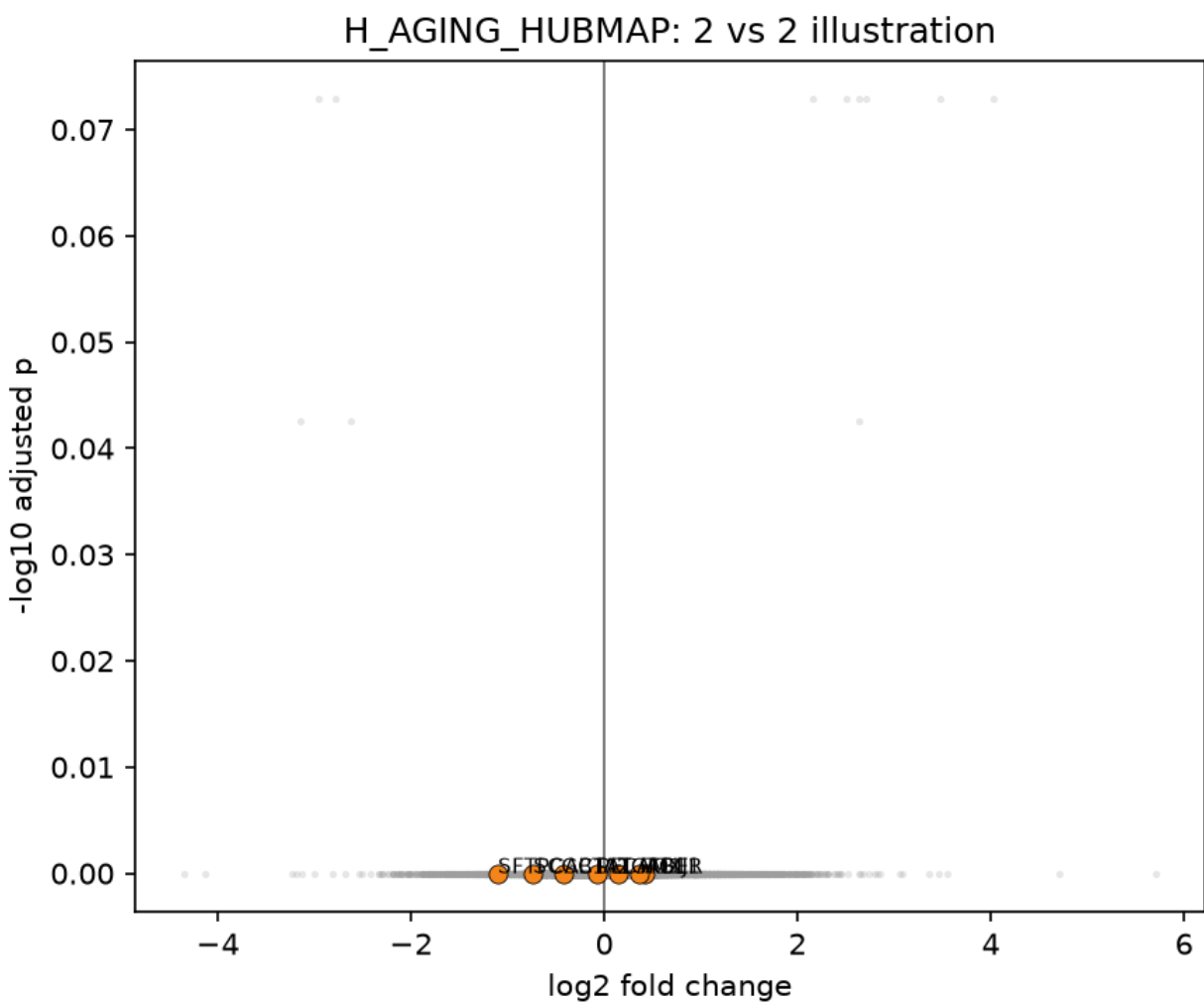
One row carries the module title. SFTPC reads about 56 in the composition-weighted HuBMAP pseudobulk (linear CP10K), about 4,156 in GTEx lung mean TPM, about 1,286,872 in GSE150910 mean control counts, and about 204,863 in OSD-248 mean ground counts for mouse [Sftpc](#). Four units and two species, with no shared scale: the spread across those columns is a units artifact, not biology. This panel is why the module compares within-resource log fold-change and pathway NES and never native levels.



Volcano plot for H_DISEASE (IPF vs control)



Volcano plot for M_FLIGHT (ISS-T flight vs ground control)



Volcano plot for *H_AGING_HUBMAP* (older vs younger), the 2 vs 2 teaching illustration

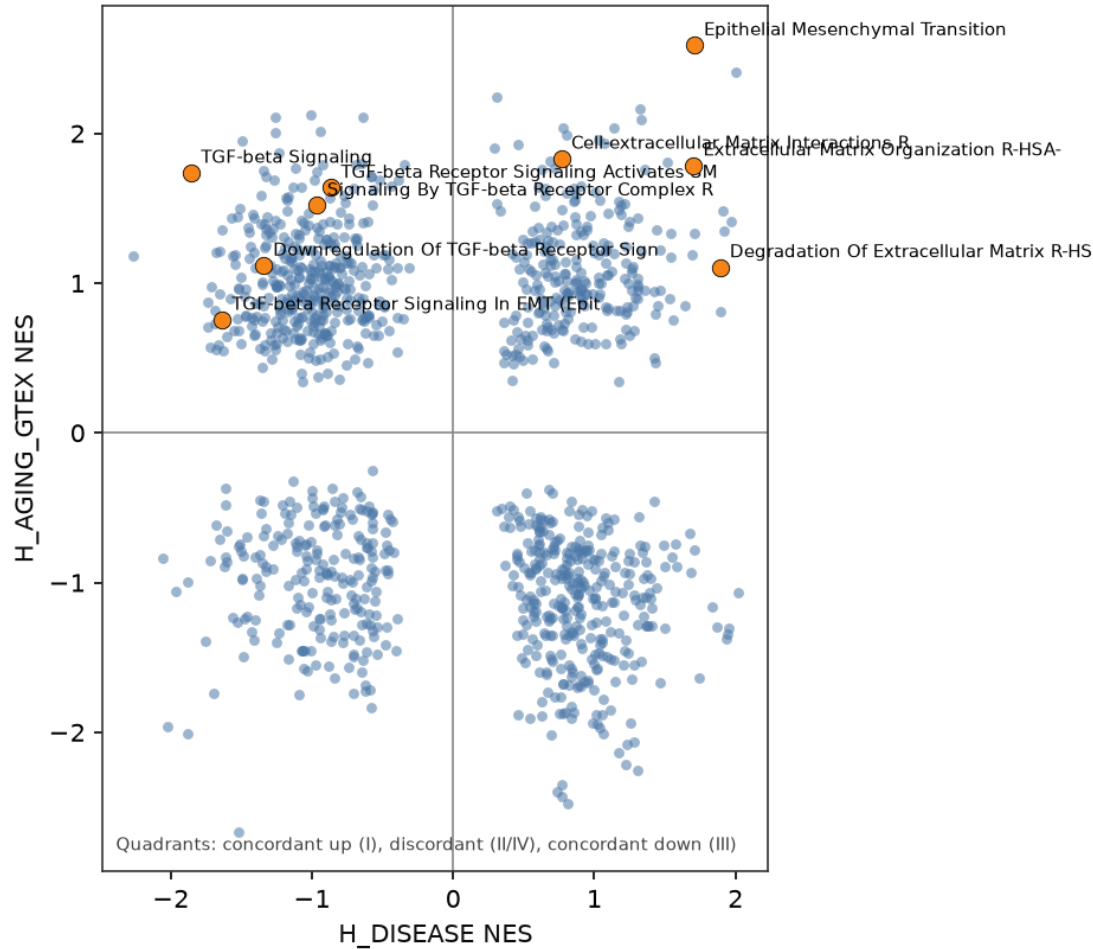
Global NES pairs (supporting)

contrast_x	contrast_y	frame	n_shared_pathways	spearman_rho	sign_agreement	spearman_p	spearman_p_note
H_DISEASE	H_AGING_GTEX	biology_best_case	1185	-0.197008	0.395781	7.82e-12	gene-set overlap violates independence; spearman_p is not a valid evidence claim for the headline
H_DISEASE	M_FLIGHT	cross_ecosystem	1136	-0.10913	0.484155	0.000229	gene-set overlap violates independence; spearman_p is not a valid evidence claim for the headline
H_AGING_GTEX	M_FLIGHT	cross_ecosystem	1158	-0.021858	0.462867	0.457419	gene-set overlap violates independence; spearman_p is not a valid

							evidence claim for the headline
H_AGING_HUBMAP	H_AGING_GTEX	method	945	0.190643	0.582011	3.48e-09	gene-set overlap violates independence; spearman_p is not a valid evidence claim for the headline

Source: module4_nes_pair_summary.tsv

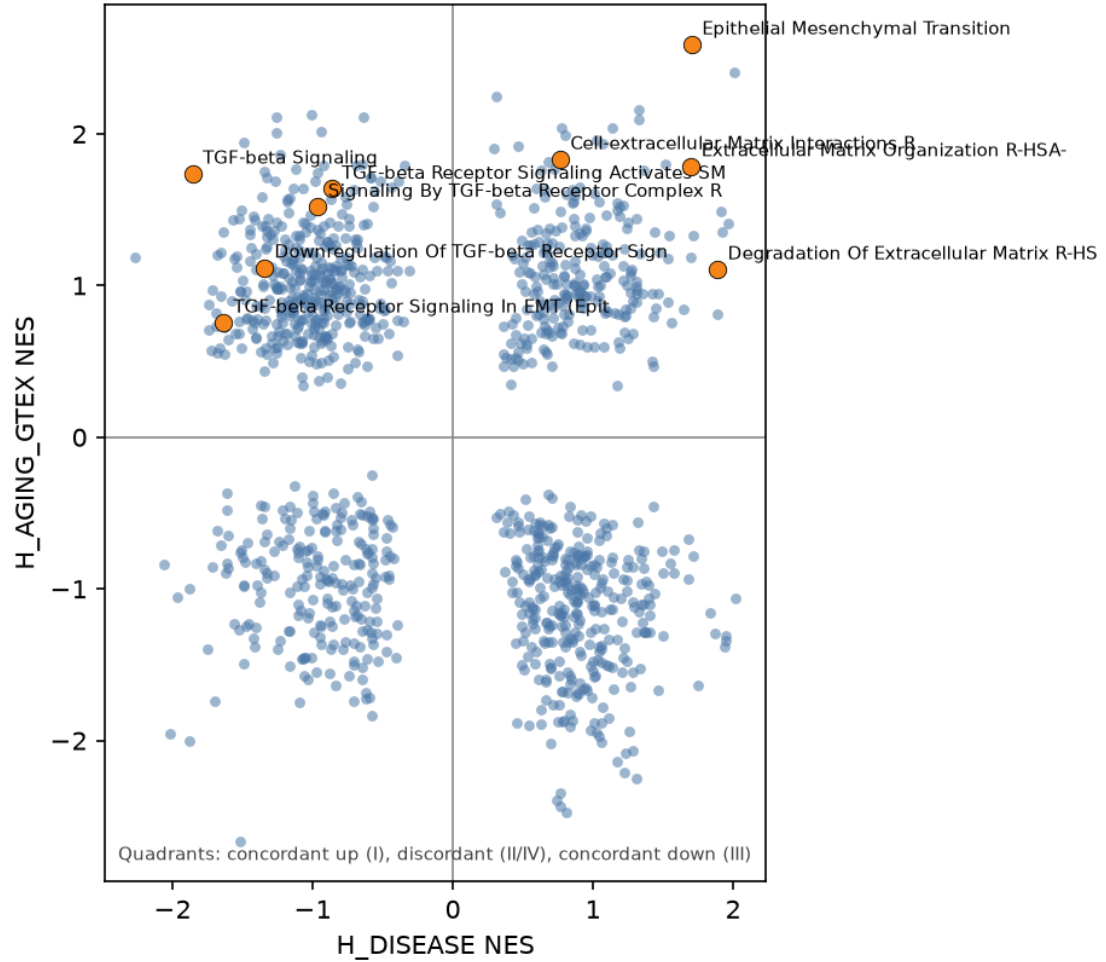
NES concordance (Hallmark + Reactome): H_DISEASE vs H_AGING_GTEX
Spearman ρ =-0.197 across 1185 shared pathways



Headline NES panel: normalized enrichment scores for H_DISEASE versus H_AGING_GTEX, with selected pathway terms labeled

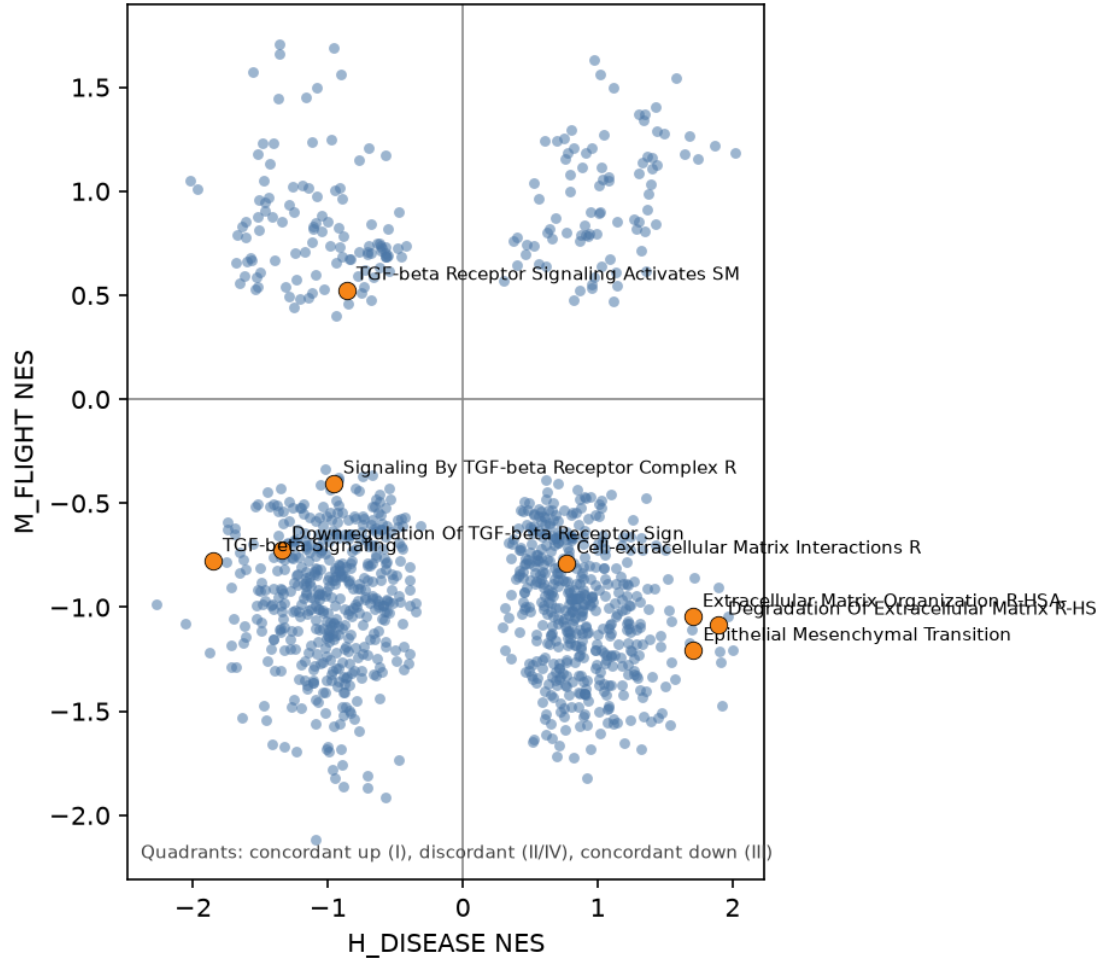
Open the panel above first. It is the canonical headline NES figure, [module4_nes_scatter.png](#), and it carries the configured headline pair [H_DISEASE](#) vs [H_AGING_GTEX](#); the explicitly named per-pair file directly below holds the same content under its pair name.

NES concordance (Hallmark + Reactome): H_DISEASE vs H_AGING_GTEX
Spearman rho=-0.197 across 1185 shared pathways



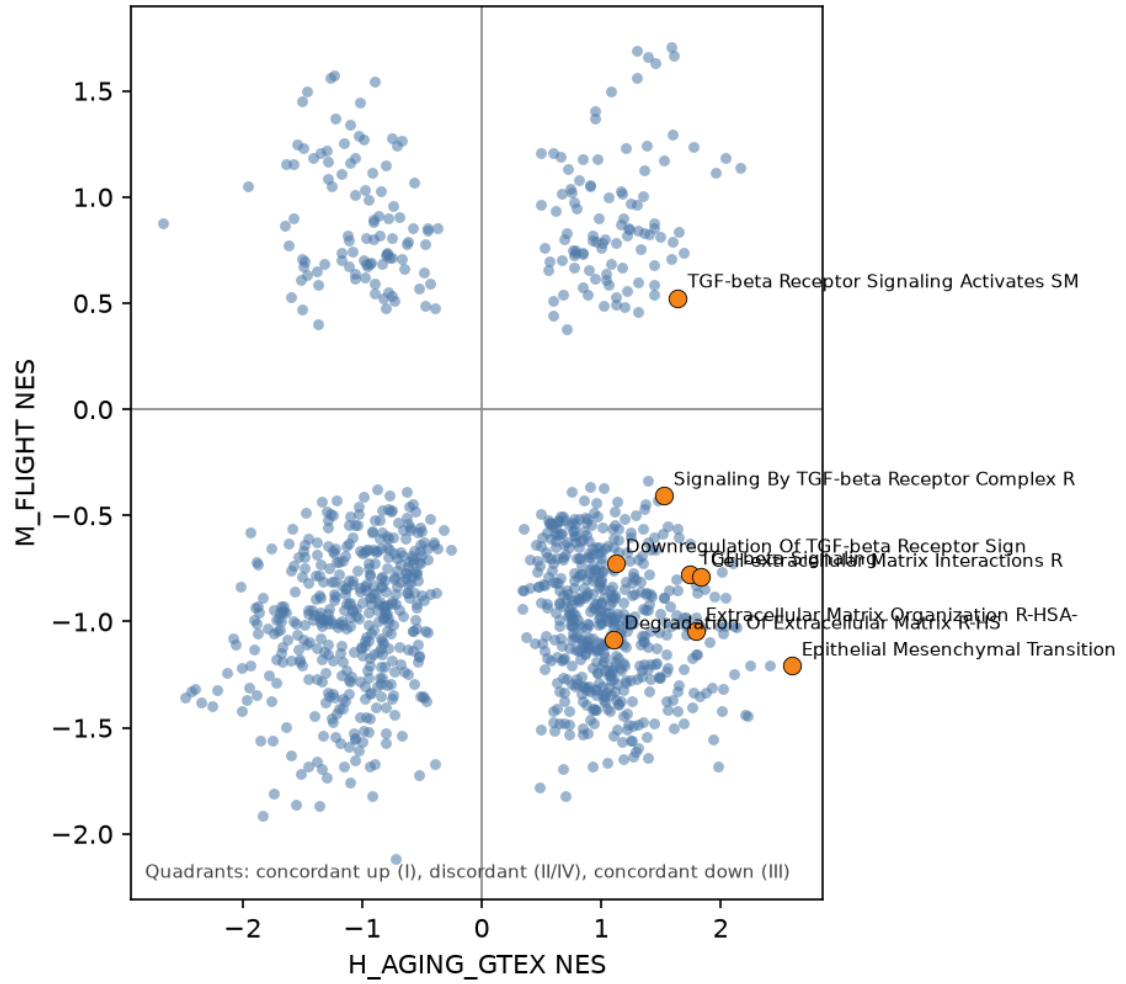
NES scatter for H_DISEASE versus H_AGING_GTEX

NES concordance (Hallmark + Reactome): H_DISEASE vs M_FLIGHT
 Spearman rho=-0.109 across 1136 shared pathways (near-zero is a result)



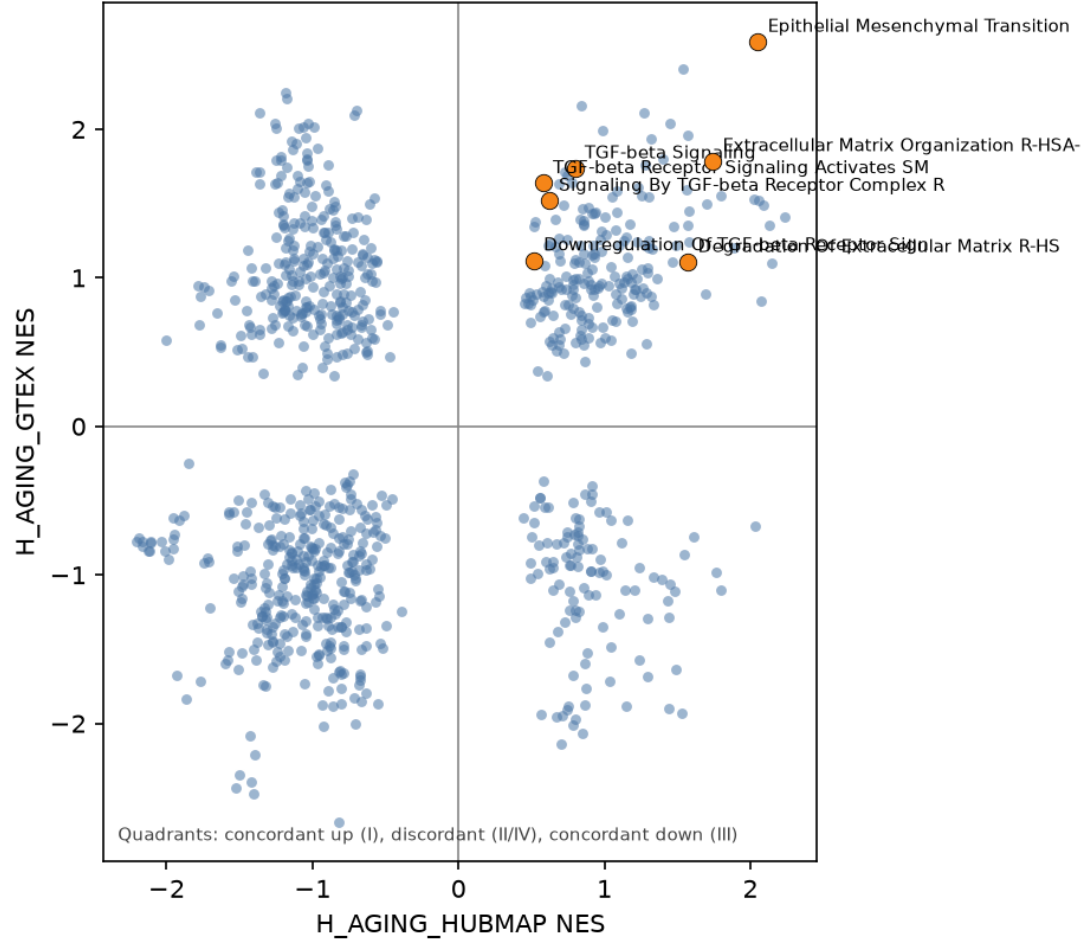
NES scatter for H_DISEASE versus M_FLIGHT

NES concordance (Hallmark + Reactome): H_AGING_GTEX vs M_FLIGHT
 Spearman rho=-0.022 across 1158 shared pathways (near-zero is a result)



NES scatter for H_AGING_GTEX versus M_FLIGHT, the aging-flight pair

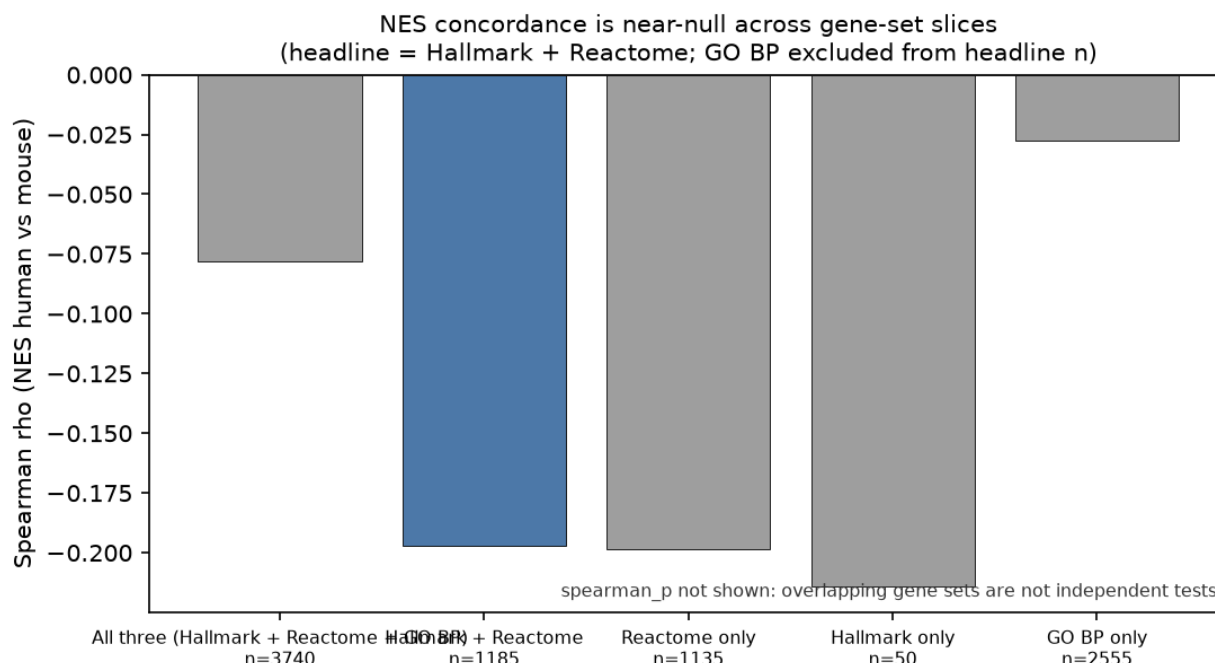
NES concordance (Hallmark + Reactome): H_AGING_HUBMAP vs H_AGING_GTEX
Spearman rho=0.191 across 945 shared pathways



NES scatter for H_AGING_HUBMAP versus H_AGING_GTEX, a method-frame pair rather than a biology pair

Biology pairs are near-null to negative on global Spearman. That is context, not the headline. Aging-flight is the flattest of the four on this axis, rho -0.022 over 1,158 shared sets, and it is still the only pair whose pathway concordance clears its permutation null; global correlation and mechanism concordance answer different questions. The fourth scatter is the [method](#) frame pair (H_AGING_HUBMAP vs H_AGING_GTEX, rho 0.191 over 945 shared sets), a pipeline check on the 2 vs 2 illustration rather than a biology result.

Gene-set library sensitivity.



Bar chart of Spearman rho for the disease-aging NES pair across five gene-set library slices

The headline coefficient moves with the library it is computed on. For [H_DISEASE](#) vs [H_AGING_GTEX](#), [module4_nes_sensitivity.tsv](#) reports Spearman rho -0.197 on the headline Hallmark plus Reactome universe (n = 1,185), -0.078 once GO BP is added to make all three libraries (n = 3,740), and -0.027 on GO BP alone (n = 2,555); Reactome alone gives -0.199 (n = 1,135) and Hallmark alone gives -0.214 (n = 50). A coefficient that ranges from -0.214 to -0.027 depending on which annotation happens to be loaded is describing the annotation at least as much as the biology. That instability, not modesty about effect size, is why the global correlation is supporting context and the permutation-tested pathway concordance is the headline.

Shared-mechanism concordance (headline)

set_id	n_mechanisms_to tal	n_mechanisms_unflag ged	n_concordant_pathw ays	n_concordant_unflag ged	perm_mea n	perm_p_ge_o bs
disease_agi ng	5	5	5	5	3.0075	0.169
disease_flg ht	0	0	0	0	0.42	1.0
aging_flight	5	3	6	3	2.483	0.0255
three_way	0	0	0	0	0.0345	1.0

Source: [module4_mechanism_summary.tsv](#)

The full null table [module4_concordance_null.tsv](#) carries 27 columns and does not fit a portrait page. The compact view below keeps the nine needed to read the close; the analytic null columns ([E_both_pass](#), [P_agree](#), [E_concordant](#), [analytic_ratio_obs_over_E](#)), the unflagged and q<0.05 concordant counts, the permutation spread ([perm_median](#), [perm_sd](#), [perm_ratio_obs_over_mean](#), [n_permutations](#)), the threshold stamps ([fdr_q](#), [nes_abs_min](#)), and the third-contrast columns ([contrast_c](#), [k_c](#), [p_c_up](#)) used by the three-way row stay in the TSV.

set_id	N_shared_pathways	k_a	k_b	p_a_up	p_b_up	observed_concordant	perm_mean	perm_p_ge_obs
disease_aging	1185	45	160	0.644	0.456	5	3.01	0.169
disease_flight	1136	38	32	0.605	0.000	0	0.42	1.000

aging_flight	1158	168	34	0.494	0.000	6	2.48	0.026
three_way	1135	38	155	0.605	0.458	0	0.03	1.000

Passer counts (k_a , k_b) are computed inside each row's own shared-pathway universe, so the same contrast can show different k across rows. On the three-way row k_a and k_b are $H_DISEASE$ and H_AGING_GTEX ; flight sits in the k_c and p_c_up columns of the TSV.

pair_id	Term	NES_a	NES_b	FDR_a	FDR_b	direction	le_jaccard	suspect_flag
disease_aging	Cholesterol Biosynthesis R-HSA-191273	-2.021249	-1.957994	0.012417	0.00959	down	0.47619	False
disease_aging	Keratinization R-HSA-6805567	2.004068	2.407191	0.0	0.0	up	0.214286	False
disease_aging	Activation Of Gene Expression By SREBF (SREBP) R-HSA-2426168	-1.878254	-2.006322	0.124168	0.004359	down	0.3125	False
disease_aging	Epithelial Mesenchymal Transition	1.706697	2.589803	0.013321	0.0	up	0.344595	False
disease_aging	Extracellular Matrix Organization R-HSA-1474244	1.701873	1.786559	0.036193	0.035369	up	0.267442	False
aging_flight	Myc Targets V2	-1.848522	-1.559738	0.0	0.049132	down	0.346939	False
aging_flight	Oxidative Phosphorylation	-1.835324	-1.915239	0.0	0.0	down	0.469512	True
aging_flight	Mitochondrial Fatty Acid Beta-Oxidation R-HSA-77289	-1.743523	-1.810352	0.044174	0.069165	down	0.423077	False
aging_flight	Respiratory Electron Transport R-HSA-611105	-1.6002	-1.628869	0.107809	0.179397	down	0.454545	True
aging_flight	Mitochondrial Protein Import R-HSA-1268020	-1.559344	-1.85967	0.121895	0.059439	down	0.480769	False
aging_flight	Respiratory Electron Transport, ATP Synthesis By Chemiosmotic Coupling, Heat Production By Uncoupling Proteins R-HSA-163200	-1.513786	-1.716195	0.147571	0.155622	down	0.425532	True

14 further columns in the source file. Source: module4_concordant_pathways.tsv

[module4_concordant_pathways.tsv](#) is the only artifact that names the pathways the close discusses. All 11 concordant rows are shown, 5 for disease-aging and 6 for aging-flight, with per-contrast NES and FDR, direction, tier, the $q < 0.05$ -in-both flag, the suspect flag, and the leading-edge Jaccard and intersection genes.

How to read the close.

- **Disease-aging: five separate concordant pathways**, grouped only by direction. Down: cholesterol biosynthesis, SREBF activation. Up: EMT, ECM organization, keratinization. Leading-edge clustering at Jaccard 0.5 merged none of them, so they are reported as five and not as blocks. The set is **not above chance**: 5 observed against a permutation null mean of 3.0, p 0.169. It does not survive as a shared-mechanism claim. Four of the five also pass $q < 0.05$ in both contrasts; SREBF activation does not.
- **ECM correction**: global correlation invited an "ECM opposite" story; the pathway screen shows ECM terms passing **up in both** IPF and aging. That is why the coefficient is not the headline.
- **Flight pairs**: pathway passers are unanimously down ($p_{up} = 0$). Treat disease-flight and three-way as **uninterpretable, not null**, while that polarity holds. Spike-in size-factor and globin accounts were tested and rejected.
- **Aging-flight: six concordant pathways, five after clustering** merged the two Respiratory Electron Transport terms. Oxidative phosphorylation, mitochondrial fatty-acid beta-oxidation and mitochondrial protein import stay distinct. Myc Targets V2 is adjacent and reported separately. The six together exceed the permutation null, 6 observed against 2.48, p 0.026, and this is the only set that does. Three of the six carry a suspect flag (oxidative phosphorylation and both Respiratory Electron Transport terms), so the result stays suspect on composition and circadian grounds.
- **HuBMAP aging**: 2 vs 2 illustration; gene-level $p_{adj} < 0.05$ count can be zero.

5. Interpretation

Module 4's teaching point is epistemic: cross-ecosystem comparison is a contrast-level problem, not a matrix merge. Methods assets can pass while biology pairs stay near-null or uninterpretable.

What the results support.

- Calibration (ERCC, GeneLab, bridge) can succeed independently of concordance claims.
- Shared-mechanism tables with permutation p -values are the correct headline objects.
- Several popular overclaims (shared fibrotic flight program; ECM opposition from rho; one aging-IPF mechanism) are contradicted by the committed tables.

Two of these results are easy to overstate, and the distance between what the tables show and what a reader will assume is where the interpretation has to be careful.

For **disease and aging**, the tables show a small set of same-sign pathway hits spanning sterol and remodeling programs, and permutation testing does not place that set above chance. The supported claim is therefore a labeled hypothesis rather than a validated shared mechanism. Neither "IPF recapitulates aging" nor "aging recapitulates fibrosis" follows from these tables alone, and the accelerated-aging framing that motivates the comparison is what makes those readings tempting enough to need stating explicitly.

For **flight**, every pathway passer moves in the same direction. Until that polarity is explained, any cross-ecosystem concordance involving flight is uninterpretable rather than null. That distinction matters in both directions: "no shared pathways" is not available as a positive biological finding while $p_{up} = 0$ dominates the passer set, because an unexplained artifact and a true absence of shared signal are not the same result.

6. Extending this analysis

- **Resolve the unanimous flight polarity first.** $p_{up} = 0$ across `M_FLIGHT` pathway passers is the one result that stalls every flight-involving pair, and the spike-in size-factor and globin accounts are already excluded. A follow-up should test the remaining candidates directly: library composition by arm (`module4_library_composition_by_arm.tsv` is already exported), RNA integrity through the RIN sensitivity contrast, and collection-time and circadian confounding, with replication in a second independent flight dataset. Until one of those explains the polarity, disease-flight and three-way stay uninterpretable rather than null.
- **Combining native levels needs a shared quantification, not a rescaling.** Any claim that merges levels across these resources would require raw reads reprocessed through one pipeline, one annotation and one unit, in one species. That work is outside this package, and the native-levels panel in Results shows what the merge would be papering over.
- **A powered HuBMAP aging statement needs a powered cohort.** `H_AGING_HUBMAP` is a 2 vs 2 illustration whose gene-level $\text{padj} < 0.05$ count can be zero; supporting an aging claim from HuBMAP would need donor arms sized like the GTEx brackets and non-overlapping age ranges, since the teaching donors sit inside the GTEx middle brackets. Treat the current contrast as a method check.
- **Lifting disease-aging above chance needs more signal or a narrower universe.** On this run 5 observed against a permutation null mean of 3.0 gives $p = 0.169$. A stronger analysis would raise per-contrast passer counts (deeper cohorts, more prerank permutations than the current 100) or pre-register a narrower pathway universe so the null is not diluted by 1,185 shared sets. Reporting the five as a labeled hypothesis is what the current numbers support.
- **Clearing the suspect flag on aging-flight needs controls, not a smaller p.** Three of the six concordant pathways are flagged (oxidative phosphorylation and both Respiratory Electron Transport terms). $p = 0.026$ does not address whether those terms track cell-composition shifts or circadian state; matched composition estimates and collection-time metadata on both sides would.
- **Stabilizing the global coefficient needs a fixed annotation.** The disease-aging rho ranges from -0.214 to -0.027 across gene-set slices, so any future use of a global NES correlation as evidence should pre-commit to one library and report the sensitivity panel alongside it.

Forward link. Module 5 (no notebook) is a scientific communication assessment on these outputs, authored separately for the course package.

7. Open questions

1. If methods calibration succeeds and biology concordance stays weak, what claim is still licensed?
2. Why is global NES Spearman the wrong headline for shared mechanisms?
3. Disease-aging gives five separate pathways, not above chance. What would promote that from hypothesis to claim?
4. Flight pathway passers are all down after ERCC and globin checks failed to explain it. What confound remains on the table?
5. Why keep flagged OxPhos / RET terms in the aging-flight set instead of dropping them?
6. What does an empty three-way intersection mean for "spaceflight as accelerated aging" rhetoric?
7. HuBMAP aging has zero genes at $\text{padj} < 0.05$. How should a learner read its NES passers?
8. When should `module4.sensitivity.enabled` be turned on, and what does it cost?

Appendix A: Analysis parameters

Live values under module4 in config/paths.yaml. Section numbers match the Methods subsections.

1. Within-resource DE (pydeseq2)

Config key	Default	Definition	If modified	Impact on results
module4.thresholds.de_min_count_sum	10	Min count sum to keep a gene	Raise	Fewer genes in each DE table
module4.geo.include_diagnoses	[ipf, control]	GSE150910 diagnoses kept	Add diagnoses	Redefines disease cohort
module4.gtex.young_age_brackets	["20-29", "30-39"]	Young arm brackets	Narrow/widen	Redefines H_AGING_GTEX
module4.gtex.old_age_brackets	["60-69", "70-79"]	Old arm brackets	Narrow/widen	Same
module4.gtex.one_sample_per_subject	true	One lung sample per subject	Set false	Pseudo-replication
module4.gtex.max_subjects_per_arm	40	Cap per arm for runtime	Raise	More power; longer DE
module4.gtex.random_seed	0	Subject subsample seed	Change	Different GTEX draws
module4.hubmap_aging.older_donors	[Donor_3, Donor_4]	Older illustration arm	Swap donors	Tiny-n DE identity changes
module4.hubmap_aging.younger_donors	[Donor_1, Donor_2]	Younger illustration arm	Swap donors	Same
module4.osd248.* sample filters	ISS-T carcass ~60d; no LAR/baseline	Flight cohort definition	Loosen filters	Different M_FLIGHT n

2. Calibration of the flight contrast

Config key	Default	Definition	If modified	Impact on results
module4.osd248.exclude_ercc_from_size_factors	true	Omit ERCC -from size factors	Set false	Spike-in mass confounds arm scaling
module4.data.ercc_exfold_expected	data/reference/ercc_exfold_expected_fc.txt	Expected Mix1/Mix 2 FC table	Replace file	Calibration slope/R2 change
module4.data.osd248_genelab_isst_de_subset	outputs/tables/module4_osd248_genelab_isst_de_subset.tsv	Committed GeneLab subset	Refresh subset	Spearman vs GeneLab changes
module4.thresholds.rin_sensitivity	6.0	QA/RIN threshold for sensitivity drop	Change	Optional filtered flight DE

module4.osd248.rin_sensitivity_threshold	6.0	Same threshold under osd248 block	Change	Same
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3. Ortholog mapping (mouse to human)

Config key	Default	Definition	If modified	Impact on results
module4.data.orthologs	data/reference/human_mouse_orthologs_one2one.tsv	One2one table path	Replace table	Loss rates and mapped DE change
module4.orthologs_meta.orthology_type	ortholog_one2one	Declared mapping class	Use many2many	Double-counting risk (not teaching default)
module4.orthologs_meta.download_date	2026-08-10	Provenance stamp	Refresh table	Should match file bytes

4. HuBMAP-GTEX level bridge (demonstration only)

Config key	Default	Definition	If modified	Impact on results
module4.data.hubmap_pseudobulk	outputs/tables/module2_composition_weighted_pseudobulk.tsv	Composition-weighted vector	Rebuild Module 2	Bridge scatter changes
module4.data.hubmap_max_across	outputs/tables/module2_max_across_labels_pseudobulk.tsv	Max-across vector	Rebuild Module 2	Same
module4.data.gtex_tpm	data/source/GTEX/gene_tpm_v10_lung.gct.gz	GTEX TPM for means	Replace release	Bridge y-axis changes

5. Pathway NES and configured pairs

Config key	Default	Definition	If modified	Impact on results
module4.enrichment.prerank.min_size	15	Min overlapping genes	Raise	Drops small sets
module4.enrichment.prerank.max_size	500	Max set size	Lower	Drops large sets
module4.enrichment.prerank.permutation_num	100	GSEA permutations	Raise	Stabler FDR; slower
module4.enrichment.comparison_gene_sets	Hallmark, Reactome	Headline NES libraries	Add GO BP	Changes shared-term universe

module4.enrichment.sensitivity_gene_sets	Hallmark, Reactome, GO BP	Sensitivity-only libraries	Edit	Sensitivity table only
module4.sensitivity.enabled	false	Recompute GO BP sensitivity	Set true	Large RAM; regenerates sensitivity TSV
module4.sensitivity.table	outputs/tables/module4_nes_sensitivity.tsv	Committed sensitivity asset	Replace file	Sensitivity panel only
module4.nes_pairs	four pairs	Which NES scatters/summaries ship	Edit	Pair tables and figures
module4.params.nes_headline_pair	[H_DISEASE, H_AGING_GTEX]	Pair attached to sensitivity narrative	Switch	Which pair sensitivity comments target

6. Shared-mechanism concordance (headline)

Config key	Default	Definition	If modified	Impact on results
module4.mechanisms.fdr_q	0.25	Passer FDR threshold	Stricter	Smaller concordance sets
module4.mechanisms.nes_abs_min	1.5	Absolute NES gate	Raise	Fewer passers
module4.mechanisms.fdr_q_strict	0.05	Stricter annotation threshold	Change	Secondary labels only
module4.mechanisms.cluster_jaccard_min	0.5	Mechanism clustering Jaccard	Lower	Larger merged clusters
module4.mechanisms.n_permutations	2000	Null permutations	Raise	Stabler permutation p
module4.mechanisms.random_seed	0	Permutation RNG	Change	Null draws differ slightly
module4.mechanisms.top_intersection_genes	40	Genes listed from LE intersection	Raise	Wider gene lists in tables

References

9. GTEx Consortium. The GTEx Consortium atlas of genetic regulatory effects across human tissues. *Science*. 2020;369(6509):1318-1330. PMID: [32913098](#).
10. Furusawa H, et al. Chronic Hypersensitivity Pneumonitis, an Interstitial Lung Disease with Distinct Molecular Signatures. *Am J Respir Crit Care Med*. 2020;202(10):1430-1444. PMID: [32602730](#). (GSE150910)
11. Love MI, Huber W, Anders S. Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. *Genome Biol*. 2014;15(12):550. PMID: [25516281](#).
12. Muzellec B, et al. PyDESeq2: a python package for bulk RNA-seq differential expression analysis. *Bioinformatics*. 2023;39(9):btad547. PMID: [37669147](#).
13. Subramanian A, et al. Gene set enrichment analysis: a knowledge-based approach for interpreting genome-wide expression profiles. *Proc Natl Acad Sci U S A*. 2005;102(43):15545-15550. PMID: [16199517](#).

14. Hayslip N, et al. Space radiation and microgravity as models of accelerated aging: modulation of hepatic miRNA-TGF-beta networks associated with senescence and fibrosis. *GeroScience*. 2026. PMID: [42337157](#).
15. Ruprecht NA, et al. Establishing a genomic radiation-age association for space exploration supplements lung disease differentiation. *Front Public Health*. 2023;11:1161124. PMID: [37250098](#).
16. NASA Open Science Data Repository / GeneLab. OSD-248 (GLDS-248), rodent spaceflight lung RNA-seq resource.